Nikhil Hanumanthu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.91
FIITJEE Miyapur	Class XII (IPE)	2024	93.50%
Delhi Public School Miyapur	Class X (CBSE)	2022	91.20%

POSITIONS OF RESPONSIBILITY

Finance & Partnerships Head | TEDxIITHyderabad

June 2026 - Present

- Lead the **Finance & Partnerships Team**, driving sponsorship outreach, pitching, negotiations, and corporate collaborations.
- Ideate and execute **new fundraising strategies** to secure financial and in-kind support for the organization.
- Manage **budgets, expenditures, and financial planning**, ensuring efficient allocation of resources across events.
- Contribute to **event strategy and planning**, coordinating partnerships, resources, and execution for TEDxIITHyderabad initiatives.

PROJECTS

Vision-Based Assistive Device for Obstacle Detection & Collision Avoidance

- Developed a wearable assistive system integrating an **ultra-wide camera**, **Raspberry Pi Zero**, **ESP32**, and **ultrasonic sensors** for obstacle perception and haptic feedback.
- Built a computer vision pipeline using **OpenCV**, **YOLOv8**, and **ByteTrack** for **object detection**, **multi-object tracking**, and **trajectory-based collision prediction**.
- Transmitted directional haptic feedback via **BLE** based on **predicted collision risk** from tracked dynamic obstacles.
- Combined vision with ultrasonic sensor fusion to detect peripheral obstacles beyond the camera's field of view, improving user safety and situational awareness.

U-Net for Skin Lesion Segmentation

- Implemented a **U-Net encoder-decoder architecture in PyTorch** for binary skin-lesion segmentation on the ISIC 2018 dataset, training on **2,594 dermatoscopic images**.
- Built an end-to-end segmentation pipeline covering **image/mask preprocessing, data loading, training, inference, and evaluation**, with Dice and IoU metrics.
- Evaluated the trained model on **1,000 test images**, achieving **82.18% Dice score and 72.20% IoU**.
- Implemented **skip connections** to preserve fine-grained spatial information lost during encoder downsampling and improve pixel-level lesion boundary prediction.

MNIST Digit Recognition Web Application

- Developed an end-to-end handwritten digit recognition application by training a Convolutional Neural Network (**CNN**) from scratch in PyTorch, achieving **99.01% test accuracy** on the MNIST dataset.
- Designed a CNN architecture with two convolutional and two fully connected layers, trained using the Adam optimizer and CrossEntropyLoss for multi-class digit classification.
- Built a FastAPI inference backend that preprocesses user-drawn digits through cropping, centering, and resizing before performing real-time prediction and confidence estimation.
- Developed an interactive HTML5 Canvas frontend enabling users to draw handwritten digits and receive **live predictions** from the deployed CNN model.

BERT Sentiment Analysis on IMDb

- Fine-tuned **BERT (bert-base-uncased)** on the **IMDb** movie review dataset (50,000 reviews) for binary sentiment classification using the Hugging Face **Trainer** API.
- Built an end-to-end NLP pipeline including data preprocessing, tokenization, dynamic padding, mixed-precision (FP16) training, and evaluation using **PyTorch**, **Transformers**, and **Datasets**.
- Achieved **92.4% test accuracy** and **0.925 F1-score**, with the trained model deployed on the **Hugging Face Hub** for

reproducible inference

SCHOLASTIC ACHIEVEMENTS

Winner (1st Place) - **Bionics Hackathon 2026, Shaastra, IIT Madras**.

Advanced through a competitive proposal selection round before presenting a working prototype during Shaastra, earning **first place** for a biologically inspired assistive technology solution.

SKILLS

Languages: Python, C++, C, SQL, HTML/CSS, JavaScript

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, XGBoost, OpenCV, YOLOv8, Pandas, NumPy, Matplotlib, SciPy

Tools: Git, GitHub, FastAPI, Jupyter Notebook, Google Colab, VS Code

RELEVANT COURSEWORK

Computer Science: Data Structures and Applications, Algorithms and Data Structures Lab, Artificial Intelligence

Mathematics : Introduction to Statistics, Introduction to Probability

EXTRACURRICULAR

- Won the **Institute Football League**, competing against teams across the campus
- Active player for the **National Sports Organisation** (NSO) football team
- Certified **Taekwondo** Black Belt (1st Dan)